

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)	(i)		$\frac{0.036 + 0.032 + 0.033 + 0.034 + 0.030}{5} = \frac{0.165}{5} = 0.033 \text{ [N]}$ Don't accept 0.03 N		1		1	1	1
		(ii)		$\frac{0.036 - 0.030}{2} = 0.003 \text{ [N]}$		1		1	1	1
		(iii)		$\frac{0.003 \text{ ecf}}{0.033 \text{ ecf}} (1)$ $\times 100 \% = 9 \% \text{ [this is less than } 10 \% \text{] so the data is repeatable (1)}$ Accept 9.09 [%] or 9.1 [%]			2	2	1	2
	(b)	(i)		Light gates are more accurate measurement or no human reaction time when using light gates Accept no human error Don't accept to make it more repeatable	1			1		1
		(ii)		Selection of 0.55 and 1.1 (1) $0.5 \times 3.3 \times 10^{-3} \times 0.55^2 = 4.99125 \times 10^{-4} \text{ [J]}$ OR $0.5 \times 3.3 \times 10^{-3} \times 1.1^2 = 1.9965 \times 10^{-3} \text{ (1) [J]}$ accept $5 \times 10^{-4} \text{ [J]}$ or $2 \times 10^{-3} \text{ [J]}$ $\frac{1.9965 \times 10^{-3}}{4.99125 \times 10^{-4}} = 4$ or $4.99125 \times 10^{-4} \times 4 = 1.9965 \times 10^{-3}$ so the student is correct (1) N.B. No penalty if $\times 10^{-3}$ is omitted throughout Alternative: $\frac{1.1^2 (1)}{0.55^2 (1)} = 4$ so the student is correct (1) or $0.3025 (1) \times 4 (1) = 1.21$ so the student is correct (1)			3	3	3	3

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	(c)	(i)		[As the drop height increases] the mean speed increases (1) at a decreasing rate (1) Don't accept speed accelerates or speed increase slows down		2		2		2
		(ii)		1.1 [m/s]		1		1		1
		(iii)	I	Weight / <u>force of</u> gravity and air resistance / drag	1			1		1
			II	Forces are balanced / equal	1			1		1
				Question 5 total	3	5	5	13	6	13